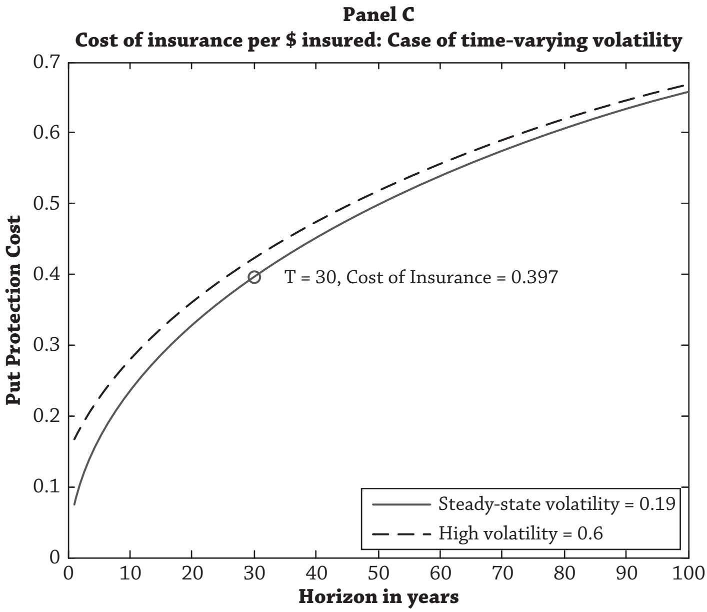

Figure 5.2 (*continued*)

than the cumulated value in bonds at the end of the period, and zero otherwise. If stocks are less risky in the long run, then the cost of this insurance should decrease over time. Panel B shows exactly the opposite: the cost of insuring against stocks underperforming bonds increases with horizon. For a thirty-year horizon, the insurance cost is 40 cents per dollar of initial investment in equities.

Bodie's analysis is clever because it doesn't matter whether stock returns are mean-reverting. One of the key insights of the Black–Scholes (1973) option pricing framework is that the expected return of stocks does not affect the value of the option.[19] Whatever the expected return of the stock, Bodie's cost of insurance in Panel B of Figure 5.2 shows that stocks are not less risky in the long run.

How can we reconcile Panel A of Figure 5.2, where the probability of stocks outperforming bonds increases over time, and Panel B, where the cost of insuring against shortfall risk also increases over time? The insurance cost in Panel B has two parts: the probability of stocks underperforming bonds and the loss contingent upon underperforming. When stocks underperform, they can really, really underperform. Investors who experienced the Great Depression or even more recently the financial crisis of 2007–2008 know that stock losses can be huge. In extreme cases you lose everything, as investors in Chinese or Russian equities learned in the early twentieth century.

[19] For a readable explanation sans differential equations, see Kritzman (2000).